DATA SCIENCE PROJECT REPORT:

(MENTAL ILLNESS)

1. PROBLEM DEFINITION:

The objective of this project is to apply data science techniques to a real-world problem. In this project, we selected **Mental illness** as our problem domain. Mental health disorders such as Depression, Anxiety, Bipolar Disorder, Schizophrenia, and PTSD are increasing globally, and early prediction can help in timely intervention.

The goal is to analyze a dataset and build machine learning models to classify whether a person is likely to have a specific mental disorder based on input features.

2. DATA COLLECTION:

The dataset used in this project was obtained from **Kaggle**. It is a real-world dataset related to mental health.

- The dataset consists entirely of **binary values (0 and 1)**.
- Each column represents either:
  - A feature (input variable), or
  - A disorder (target variable)

Dataset Link: https://www.kaggle.com/datasets/karanbakshi1/mental-illness-dataset

3. DATA PREPROCESSING:

The dataset is suitable for classification tasks and contains sufficient features for analysis.

Since the dataset is already in binary format, preprocessing was relatively straightforward:

- **Missing Values:** Checked and handled using appropriate methods.
- **Encoding:** Not required because data is already binary.
- **Outliers:** Not significant due to the binary nature of data.
- **Feature Scaling:** Not required for binary data.
- **Train-Test Split:** Data was split into training and testing sets (80% training, 20% testing).

4. **EDA (EXPLORATORY DATA ANALYSIS):**

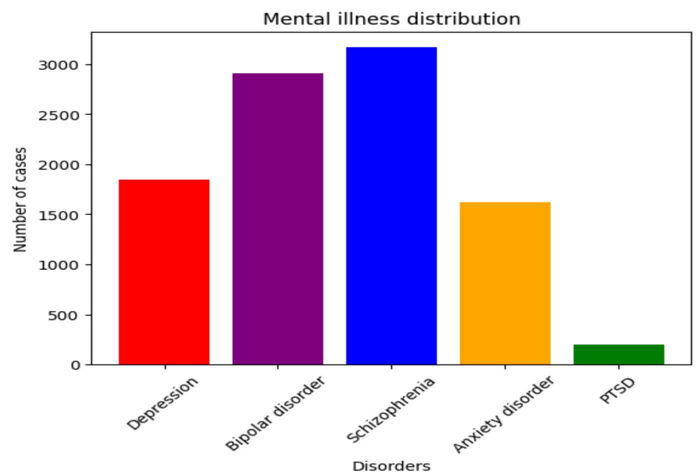
EDA was performed to understand the structure and patterns in the dataset.

Techniques Used:

- Bar graphs for distribution of disorders
- Heatmap for correlation analysis

Graphs and charts were used to visually understand the dataset.

A bar graph was created to show the distribution of mental illnesses using the sum of binary values (0/1) for each disorder. The graph compares the number of cases for Depression, Bipolar Disorder, Schizophrenia, Anxiety Disorder, and PTSD. Different colors were used for each disorder to improve visualization and clarity. This graph helps in identifying which disorder is most common and which is least frequent in the dataset.



5. **FEATURE ENGINEERING:**

Feature engineering was done by converting binary symptom data (0/1) into meaningful features. Symptoms were grouped by disorder, and scores for Bipolar Disorder, Depression, Anxiety, Schizophrenia, and PTSD were created by counting how many symptoms are present. A combined feature **Total Mental Health** was also created to show overall symptom load. Feature selection was based on correlation and importance, keeping only the most useful features and reducing unnecessary ones. Since the data was already binary, no major transformation was needed, but creating scores helped make the data more balanced and easier for models like Logistic Regression and Random Forest to understand and predict better. This step helped in improving model performance and reducing complexity.

6. **MODEL BUILDING**:

The problem is considered a classification task because the target variables are binary (0 and 1), representing the presence or absence of mental disorders. As the outcome falls into categories instead of numerical ranges, classification models are used.

Two machine learning models were implemented:

1. **Logistic Regression**
2. **Random Forest**

These models were chosen because the problem is a classification task.

- Logistic Regression works well with binary data.
- Random Forest helps capture complex patterns.

7. MODEL EVALUATION:

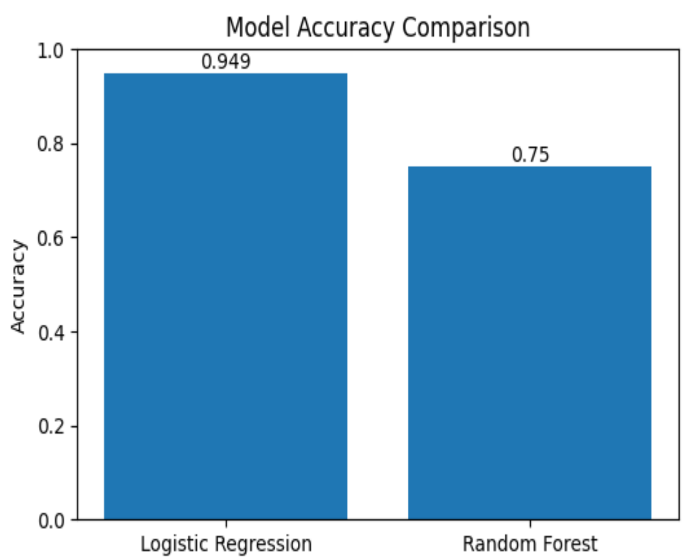
Models were evaluated based on:

- Accuracy Score
- Confusion Matrix

Accuracy score and confusion matrix were used for model evaluation because they are appropriate for classification tasks. Accuracy shows the overall correctness of the model, whereas the confusion matrix provides deeper insight by showing how many predictions were correct and where the model made mistakes (such as false positives and false negatives).

Results:

- Logistic Regression showed better performance compared to Random Forest.
- It achieved higher accuracy and produced more reliable classification results.
- Random Forest performed well but was slightly less accurate.



8. WEBSITE:

A web-based **Mental Health Prediction System** was developed to provide an interactive platform for predicting mental health conditions. The frontend was built using HTML, CSS, and JavaScript, while the backend was implemented using Flask.

The system allows users to select up to atmost seven symptoms from a predefined list related to disorders such as depression, anxiety, bipolar disorder, schizophrenia, and PTSD. After submission, the selected data is sent to the backend via a POST request, where it is compared with predefined symptom sets. A match score is calculated for each disorder, and the condition with the highest score is returned as the predicted result. If no strong match is found, the system indicates that no clear disorder is detected.

The result is displayed dynamically on the frontend with features like typing animation, dark mode, and interactive selection, enhancing user experience. Overall, the website demonstrates a simple rule-based prediction system combined with a clean and user-friendly interface for mental health awareness.

9. CONCLUSION:

In conclusion, this project successfully applied data science techniques to the domain of mental health prediction. Starting from data collection through Kaggle, the dataset was preprocessed and analyzed using EDA to understand patterns and relationships. Feature engineering further improved the dataset by creating meaningful features such as disorder scores and overall mental health indicators.

Machine learning models, including Logistic Regression and Random Forest, were then implemented for classification. Based on evaluation using accuracy score and confusion matrix, Logistic Regression showed better performance for this binary dataset. Additionally, a web-based system was developed to make the prediction process interactive and user-friendly.

However, the dataset's binary nature limits deeper analysis and may not fully reflect real-world scenarios. In the future, using more complex data and advanced models can improve prediction accuracy. Overall, this project highlights the importance of early detection and demonstrates how machine learning can support mental health analysis and awareness.

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